

DINOv3 embeddings for Hierarchical Concept-Based Classification

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Abstract

We present a hierarchical concept-based classification framework built on top of frozen DINOv3 embeddings. Instead of assigning each input to a single leaf label, the model traverses a semantic hierarchy top-down and returns the most specific reliable prediction, stopping at an internal ancestor node when the visual evidence is insufficient. Each node of the hierarchy is modeled as an uncertainty-aware concept region using Extreme Value Machines (EVMs), which provide a principled open-set rejection mechanism without requiring any fine-tuning of the vision backbone. We evaluate the framework on COCO and iNat21-Mini, showing that the hierarchical EVM surpasses flat KNN and logistic regression baselines in hierarchical accuracy once the rejection mechanism is activated, while retaining coverage above 70% on COCO and 65% on iNat21-Mini. Ablation studies confirm that sibling negatives are sufficient to train discriminative node-level EVMs at less than half the fitting cost of full-memory negatives, and that FAISS-based negative selection consistently outperforms random sampling.

1. Introduction

Self-supervised learning has become a central paradigm in computer vision, enabling models to learn rich visual representations without relying on large amounts of manually annotated data. Self-supervised vision transformers, such as DINO [4], have shown that semantic structure can emerge directly from visual data, yielding embeddings that transfer effectively to downstream tasks such as classification, retrieval, and localization. In parallel, latent-space predictive methods such as I-JEPA [1] further highlight the importance of learning abstract semantic representations rather than reconstructing low-level image details.

Recent vision foundation models further strengthen this trend. Models such as DINOv3 [22] provide powerful feature extractors whose frozen representations can be reused across a wide range of visual recognition tasks. Nevertheless, downstream classifiers built on top of these embed-

dings are typically flat: they assign each input to a single target class, without explicitly modeling the hierarchical structure of visual categories or the semantic reasoning path that leads to the prediction.

This limitation is particularly relevant in fine-grained recognition scenarios, where different errors should not be treated equally. For instance, confusing two visually similar subclasses is less severe than predicting a semantically distant category. Nevertheless, standard classifiers usually ignore this structure and evaluate predictions only at the final label level.

A natural way to address this limitation is to exploit hierarchical structure. Instead of representing classification as a flat decision among leaf labels, labels can be organized into a tree of increasingly specific concepts. Under this formulation, a prediction is not only evaluated at the leaf level, but also through its ancestors.

A closely related framework is proposed in *Egocentric Hierarchical Visual Semantics* [7], where egocentric visual encounters are used to incrementally build a hierarchy of visual concepts. In that framework, each encounter is embedded into a visual feature space and compared with previously observed instances stored in memory, enabling an online, instance-based form of object recognition. The hierarchy is updated through user feedback, which guides the placement of new encounters at the appropriate level of semantic specificity. However this method is designed for videos and only takes into account an online setting, which is inefficient on larger datasets.

Image hierarchical classifiers address part of this issue by organizing labels at multiple levels of abstraction [2, 24]; however, they often assume a predefined hierarchy with a fixed set of detail levels.

We argue that a hierarchical representation of concepts can make predictions both more flexible and more interpretable. If the model is uncertain about the most specific class, it should be able to stop at an ancestor node that is still semantically meaningful. This allows ancestor predictions to be considered partially correct and provides hints about the visual concepts that support the decision.

In this work, we investigate whether embeddings from

frozen vision foundation models can be organized into an interpretable hierarchy of visual concepts. We represent the classification space as a concept tree, where each node corresponds to a semantic concept at a specific level of abstraction. Given an input image, we extract visual embeddings using a frozen foundation model and evaluate them against this hierarchy: the prediction is formulated as the selection of the most specific reliable node, rather than always forcing a decision at the leaf level, so that ancestor predictions can be considered meaningful and partially correct rather than simply wrong.

The difficulty lies in requiring the model to be *well-behaved*: confident only when the evidence truly supports it, and able to *reject* a sample at a node rather than forcing a choice among its children, as a flat classifier would under the assumption of an i.i.d. label space. This is why we rely on Extreme Value Machines (EVMs) [21], which model class boundaries explicitly rather than relying on a single confidence score.

We do not aim to fine-tune the vision foundation model itself. Instead, we focus on using its frozen embeddings as a representation space in which hierarchical concepts can be modeled and evaluated.

The main contributions of this work are as follows:

- We propose a hierarchical concept-based classification framework built on top of frozen vision foundation model embeddings.
- We model each node of the hierarchy as an uncertainty-aware concept region using Extreme Value Theory.
- We introduce a prediction mechanism that can return either a fine-grained leaf label or a more general ancestor when the evidence is insufficient.
- We propose a hierarchy-agnostic formulation that does not require a fixed number of levels or level-specific prediction heads, making it applicable to arbitrary tree structures.

2. Related Works

Existing flat classifiers [6, 11] force the model to always return a leaf prediction, even when the visual evidence may only support a more general concept. This can lead to overconfident fine-grained predictions and provides limited insight into why a certain label was selected.

Concept-based methods aim to explain model predictions using human-interpretable intermediate variables. Concept Bottleneck Models introduce an intermediate concept layer and predict the final class from concept activations, enabling interpretation and human intervention at test time [15]. Related approaches such as Concept Activation Vectors [14] represent concepts as directions in the latent space of a trained model and measure their influence on predictions. These methods provide interpretable explanations, but they often require explicit concept annotations or treat

concepts as a flat set of independent attributes. In contrast, we investigate whether concepts can be organized hierarchically in the embedding space of frozen vision foundation models.

Hierarchical classification exploits taxonomies to organize labels at multiple levels of abstraction, allowing predictions to be evaluated not only at the leaf level but also according to their semantic distance in the hierarchy. Large-scale datasets such as ImageNet are naturally organized according to the WordNet hierarchy [5], while fine-grained datasets such as iNaturalist provide biological taxonomies relating species, genera, families, and higher-level categories [12].

Several approaches have incorporated hierarchical information into image classification. HD-CNN [24] introduces a coarse-to-fine architecture in which a coarse classifier routes samples to specialized fine-grained classifiers. Other methods modify the training objective instead of the architecture. For example, *Making Better Mistakes* [2] uses hierarchy-aware losses and soft labels to reduce the semantic severity of classification errors. More recent work on hierarchical selective classification [9] allows pretrained classifiers to retreat to internal nodes under uncertainty, studying the trade-off between prediction specificity and hierarchical risk. Visually consistent hierarchical classification methods, such as H-CAST [19], enforce consistency across hierarchy levels by aligning coarse and fine predictions through visual grounding.

Self-supervised vision transformers have shown that strong semantic representations can emerge without manual labels. DINO demonstrated that self-supervised ViT features contain explicit information about semantic object structure and perform well under k-NN and linear evaluation [4]. More recent foundation models, such as DINOv2 and DINOv3, further improve the quality and robustness of frozen visual features across image and pixel-level tasks [18, 22]. However, while these foundation models remove the need for manually annotated training data, they do not address the limitations of flat classifiers discussed above: downstream uses still typically pair these embeddings with a single-label, flat prediction head [10, 20].

Standard closed-set classifiers are forced to assign each input to one of the known classes, even when the visual evidence is insufficient or the sample lies outside the training distribution. Open-set recognition addresses this limitation by allowing models to reject unknown or unreliable predictions. The Extreme Value Machine further models class boundaries using Extreme Value Theory, providing a principled mechanism for open-set recognition and rejection [21]. Our work builds on this perspective, but applies EVT-based modeling at the level of nodes in a semantic hierarchy, allowing the classifier to stop at the most specific reliable concept rather than always forcing a leaf prediction.

Our work differs from these approaches in that we do *not* train a new hierarchical backbone, modify the loss of the vision model, or train each layer to represent a specific fine-grained level. Instead, we use frozen DINOv3 embeddings as a fixed representation space and model each node of the hierarchy as an uncertainty-aware concept region. This makes the method independent of a fixed number of hierarchy levels: the same node-level mechanism can be applied to arbitrary tree structures, including unbalanced hierarchies with variable depth and branching factors.

3. Method

The proposed pipeline takes an image as input and returns the most specific reliable node in a semantic hierarchy. The framework is inspired by *Egocentric Hierarchical Visual Semantics* [7] and combines a frozen DINOv3 encoder with a static hierarchy of concept classifiers.

3.1. Overview

We address hierarchical image classification in a static hierarchy setting. Given an input image, we first extract a visual embedding using a frozen DINOv3 backbone. The resulting embedding is then classified through a fixed semantic hierarchy, where each internal node is associated with a set of Extreme Value Machine (EVM) classifiers over its children. Instead of directly assigning each sample to one of the leaf classes, the model traverses the hierarchy from coarse to fine concepts and returns the most specific reliable prediction, which may be either a leaf node or an internal ancestor when the visual evidence supports only a coarser concept. The predicted path from root to final node provides a natural concept-level explanation of the decision.

3.2. Visual Embedding Extraction

For each image crop, we extract a global representation using the CLS token of the frozen DINOv3 ViT-7B/16 encoder. Since the fitting procedure of the downstream classifiers scales with the dimensionality of the input features, we apply a Johnson–Lindenstrauss random projection [8] to obtain a lower-dimensional representation. In particular, the original 4096-dimensional embeddings are projected to 1024 dimensions, reducing the computational cost while approximately preserving pairwise distance of the embedding space with high probability: a random linear projection concentrates vector norms around their expectation, so the projection error on any pairwise distance can be made small with probability depending only on the number of points, not the original dimensionality.

Let f denote the frozen encoder and $R \in \mathbb{R}^{1024 \times 4096}$ the Johnson–Lindenstrauss random projection matrix. Given an image crop I , the embedding is computed as

$$x = R f_{\text{CLS}}(I) \in \mathbb{R}^{1024}.$$

The encoder is never fine-tuned. All classifiers are trained on top of these fixed embeddings.

3.3. Static Hierarchy Construction

The semantic hierarchy is given and can vary depending on the domain we want to classify. The tree is defined as follows:

$$\mathcal{T} = (\mathcal{V}, \mathcal{E}),$$

where each node $v \in \mathcal{V}$ represents a semantic concept and each edge $(v, u) \in \mathcal{E}$ represents a parent-child relation.

The hierarchy is **static**: its topology is initialized once and is not modified during training. Therefore, no new nodes or edges are added by the learning procedure.

3.4. Node-Level EVM Classifiers

Each internal node v defines a local classification problem over its children. Let $\text{Ch}(v)$ denote the set of child nodes of v . For each child $u \in \text{Ch}(v)$, we train an EVM model E_u that estimates whether an embedding belongs to the subtree rooted at u .

The positive set for child u is defined as the set of embeddings whose labels belong to the subtree of u :

$$X_u^+ = \{x_i : y_i \in \text{subtree}(u)\}.$$

The negative set is constructed from sibling subtrees:

$$X_u^- = \{x_i : y_i \in \text{subtree}(w), w \in \text{Ch}(v), w \neq u\}.$$

Thus, each node learns to discriminate among its children using only local sibling information.

Negative Selection In practice, the full negative set X_u^- can be large; we therefore consider different policies for subsampling it. A random policy samples a subset of sibling negatives, while a nearest-neighbor policy retrieves the closest (based on *cosine similarity*) sibling negatives for each positive example using FAISS [13]. If no subsampling is applied, all sibling negatives are used.

This design allows us to test whether the quality of negative examples affects the learned EVM boundaries and the resulting hierarchical predictions.

EVM Fitting For each child node, the EVM is fitted using its positive embeddings and the selected sibling negatives. For every positive embedding, distances to negative examples are computed and a Weibull distribution is fitted on the nearest negative distances. This allows the EVM to estimate a membership probability for the corresponding concept region.

3.5. Hierarchical Inference

At inference time, classification proceeds top-down through the hierarchy. Starting from the root node, the model evaluates the EVM membership scores of all children of the current node. The child with the highest score is selected only if its score exceeds the open-set threshold λ ; otherwise, the prediction stops at the current node.

Formally, at node v , the selected child is

$$u^* = \arg \max_{u \in \text{Ch}(v)} E_u(x), p^* = E_{u^*}(x)$$

The transition rule is

$$\begin{cases} v \leftarrow u^*, & \text{if } p^* \geq \lambda, \\ \text{stop at } v, & \text{otherwise.} \end{cases}$$

The final output is therefore either a leaf node or an internal ancestor node, depending on the confidence of the hierarchical prediction. The procedure is summarized in Algorithm 1.

Algorithm 1 Node-Level Classification

Require: Embedding x , root node r , hierarchy $\mathcal{T}$, EVM models $\{E_u\}$, threshold λ

Ensure: Most specific reliable node $\hat{v}$

```

1:  $v \leftarrow r$ 
2: while  $v$  is not a leaf do
3:   Let  $\text{Ch}(v)$  be the set of child nodes of  $v$ 
4:   for all child nodes  $u \in \text{Ch}(v)$  do
5:      $p_u \leftarrow E_u(x)$ 
6:   end for
7:    $u^* \leftarrow \arg \max_{u \in \text{Ch}(v)} p_u$ 
8:    $p^* \leftarrow p_{u^*}$ 
9:   if  $p^* \geq \lambda$  then
10:     $v \leftarrow u^*$ 
11:   else
12:     break
13:   end if
14: end while
15: return  $\hat{v} \leftarrow v$ 

```

4. Experiments

We conduct a set of experiments to evaluate the proposed hierarchical classification pipeline. We first assess whether DINOv3 embeddings provide a suitable representation space for the task (Section 4.1). We then evaluate the impact of the hierarchy by comparing the proposed formulation against non-hierarchical and flattened alternatives (Section 4.2), and analyze whether negative examples can be restricted to sibling nodes (Section 4.3). We then compare different strategies for selecting negative examples during EVM updates (Section 4.4). Finally, we study the effect

of the Johnson–Lindenstrauss target dimension on classification performance (Section 4.5).

Unless otherwise specified, all experiments in this section are conducted directly on the full 4096-dimensional DINOv3 embeddings, without applying the Johnson–Lindenstrauss projection, in order to isolate the effect of each design choice from the additional variance introduced by the projection itself. The effect of the projection is studied separately in Section 4.5, and the resulting target dimension $k=1024$ is used only in the final evaluation (Section 7.4).

4.1. DINOv3 Feasibility

We conducted a preliminary experiment to assess whether a DINO encoder could be integrated into the pipeline proposed in *Egocentric Hierarchical Visual Semantics* [7]. The original method relies on an unsupervised CNN-based feature extractor, derived from the DeeperCluster approach [3], to compute visual object embeddings.

In our experiment, we replaced the original visual embedding backbone with DINOv3, while keeping all downstream components of the pipeline unchanged. The goal was to evaluate whether DINOv3’s self-supervised visual representations could provide more discriminative features than the original encoder and whether they could be used as a suitable input representation for the hierarchical EVM-based classifier. Results for this experiment are reported in Appendix 11.1.

4.2. Hierarchy ablations

To assess the role of the hierarchy, we evaluate alternative structural configurations while keeping the visual representation fixed.

- **No hierarchy.** We evaluate a non-hierarchical formulation based on independent EVM classifiers, where the negative set for each class is composed of all other (adversarial) classes. This setting can be interpreted as an EVM-based classification baseline and is used to assess whether EVMs are suitable for the proposed pipeline independently of the hierarchical structure.
- **Flat hierarchy.** We evaluate a simplified hierarchical formulation composed of a single root node and one leaf node for each dataset class. This setting preserves the EVM-based prediction mechanism while removing intermediate semantic levels, allowing us to isolate the effect of hierarchy depth and structure.

Results for this ablation are reported in Appendix 11.2.

4.3. Sibling negative ablation

We investigate whether the negative examples used to update each node can be restricted to sibling nodes rather than being sampled from the entire hierarchy. Since the hierarchy is fixed during training, sibling nodes represent the

most semantically relevant competing concepts for a given node. This ablation is intended to evaluate whether sibling-based negatives are sufficient to learn discriminative node-level EVMs, while reducing the number of negative examples considered during updates.

4.4. Negative selection ablation

We compare different strategies for selecting negative examples during EVM updates. In particular, we consider:

- **All negatives.** All examples contained in the sibling nodes are used. This setting provides the most complete negative set, but it is also the most computationally expensive.
- **Random sampling.** A random subset of negative examples is selected. This baseline is used to assess whether random sampling can approximate the behavior of using the full negative memory at a lower computational cost.
- **FAISS-based selection.** Negative examples are selected through nearest-neighbor retrieval in the embedding space using FAISS. This strategy aims to identify informative negatives while keeping the retrieval procedure efficient and more precise than sampling.

4.5. JL Dimensionality Ablation

We evaluate the effect of the Johnson–Lindenstrauss target dimension k on classification performance, comparing $k \in \{512, 1024, 2048, 4096\}$ against no reduction, keeping the visual representation and hierarchy fixed. This ablation is intended to empirically validate the distortion/dimension trade-off discussed in Appendix 11.2 and to motivate the choice of $k = 1024$ used in the main experiments.

5. Evaluation

We describe the metrics used to assess the proposed hierarchical classifier and the dataset on which all experiments are conducted.

5.1. Evaluation metrics

Since the proposed method does not always force a prediction at the leaf level, standard flat accuracy is not sufficient to evaluate the model. The classifier may stop at an internal node when the visual evidence is not reliable enough to support a more specific prediction. Therefore, we evaluate both the correctness and the specificity of the predicted node.

Hierarchical correctness (HAcc). Given a ground-truth leaf label y and a predicted node $\hat{y}$, we consider the prediction hierarchically correct if $\hat{y}$ is either equal to y or belongs to the set of ancestors of y :

$$\text{correct}_H(\hat{y}, y) = \mathbb{I}[\hat{y} \in \text{Anc}(y) \cup \{y\}].$$

This allows ancestor predictions to be treated as correct but less specific decisions.

Mean Geodesic Distance (MGD). To measure the semantic severity of predictions across the test set, we compute the mean geodesic distance between predicted and ground-truth nodes in the hierarchy, where the geodesic distance corresponds to the length of the shortest path connecting two nodes:

$$d_G(\hat{y}, y) = \text{shortest_path}(\hat{y}, y), \quad \text{MGD} = \frac{1}{N} \sum_{i=1}^N d_G(\hat{y}_i, y_i).$$

Coverage (Cov). Since the model can stop at internal nodes, we measure how informative each prediction is. Instead of using normalized depth, which can be misleading in unbalanced hierarchies, we define coverage based on the number of leaf nodes contained in the predicted subtree:

$$\text{coverage}(\hat{y}) = 1 - \frac{\log |\text{Leaves}(\hat{y})|}{\log |\text{Leaves}(r)|},$$

where r is the root node and $\text{Leaves}(\hat{y})$ is the set of leaf descendants of the predicted node. This assigns zero coverage to the root and maximum coverage to leaf predictions. Internal nodes receive intermediate values depending on how much uncertainty remains under the predicted concept.

5.2. Dataset

We evaluate the proposed method on the COCO dataset [16]. COCO is chosen because it provides a diverse collection of real-world images containing multiple object categories, making it a useful benchmark for testing whether visual embeddings can support concept-level recognition across high-level classes.

However, the original COCO dataset is not directly designed for single-label hierarchical classification. First, COCO images often contain multiple objects belonging to different categories. To adapt the dataset to our setting, we preprocess the images by extracting individual object crops using the provided annotations; crops that are too small or otherwise unsuitable for feature extraction are discarded. The resulting number of images are $\simeq 800\text{k}$.

Second, COCO provides a flat set of object labels rather than a semantic hierarchy. To obtain hierarchical supervision, we rely on prior mappings [23] between COCO categories and WordNet synsets [17]. The resulting WordNet subset is then used to define parent-child relations between concepts and to organize the original flat labels into a hierarchy.

A more common benchmark throughout the hierarchical classification literature is iNaturalist, and in particular its Mini version, iNat21-Mini [12]. iNat21-Mini contains 10,000 species sampled from the full iNat2021 training set, with 50 training images per species, and organizes them into a 7-level biological taxonomy (kingdom to species).

This balanced, moderate-scale subset is widely adopted as a standard fine-grained benchmark for hierarchical classification and selective classification methods [9], and we adopt it here to evaluate our method under a more fine-grained hierarchy than COCO.

6. Implementation Details

All experiments share the same frozen visual representation. Image embeddings are extracted using the DINOv3 ViT-7B/16 encoder and kept fixed throughout training and evaluation. A Johnson–Lindenstrauss random projection (seed=0) is applied to reduce the embedding dimensionality from 4096 to 1024 before fitting any classifier. The dataset is split into 90% training and 10% test samples (seed=0). The distance metric used for all experiments is *cosine similarity*. The EVMs are fitted using a *tail size* of 25. All experiments were run on a single NVIDIA A100 GPU.

6.1. Ablation Experiments

The hierarchy ablations (Section 4.2), sibling negative ablation (Section 4.3), and negative selection ablation (Section 4.4) are conducted on the COCO validation set preprocessed as described in Section 5.2. In these experiments the threshold is fixed at $\lambda = 0$, so the model always predicts a leaf node. The reported metrics are intended to isolate the effect of each design choice and should not be interpreted as final system performance.

6.2. Evaluation

For the final evaluation, the full training set is used to fit the system. We compare the proposed hierarchical EVM formulation against two flat classification baselines trained on the same frozen embeddings.

Logistic regression baseline. A logistic regression head is trained on top of the frozen DINOv3 embeddings to predict the target class directly. This serves as a strong flat classification baseline and measures how linearly separable the embeddings are without any hierarchical structure.

KNN baseline. A k -nearest neighbour classifier ($k = 5$, cosine distance, distance-weighted) is used as a second flat baseline, following common practice for evaluating frozen foundation model embeddings.

Complete hierarchy. The full hierarchy is used to fit EVM classifiers at each internal node, using FAISS-based negative selection. This corresponds to the complete hierarchical formulation proposed in this work.

7. Results

7.1. Siblings Negative Ablation

We investigate whether restricting the negative examples to sibling nodes is sufficient to learn discriminative EVM boundaries, while reducing the computational cost of the fitting procedure. Table 1 compares the use of all negatives stored in memory against the use of only sibling negatives for each node.

Table 1. Comparison between all-memory negatives and sibling negatives. Tested over 5 random seeds.

Neg. set	Acc. (%) $\uparrow$	Geo. Dist. $\downarrow$	Fit time (rel.) $\downarrow$
All mem.	68.07 ± 0.56	1.81 ± 0.74	$1.00\times$
Siblings	68.08 ± 0.76	1.78 ± 0.03	$0.32\times$

The two strategies achieve the same accuracy and geodesic distance, suggesting that sibling negatives provide a good approximation of the full-memory setting in this experiment. At the same time, restricting the negative set to siblings substantially reduces the fitting time: the absolute fit time drops from 702.8s using all negatives to 223.5s using only siblings, requiring less than a third of the time needed by the full-memory setting.

7.2. Negative Selection Ablation

This experiment tests how results change using FAISS or random sampling for negative selection. It has been tested only with one FAISS configuration but 5 runs with different seeds for each random value tested.

Table 2. Comparison of negative selection strategies. Tested over 5 random seeds.

Strategy	Neg. size	Acc. (%) $\uparrow$	F1 (%) $\uparrow$	MGD $\downarrow$
Random	10%	61.92 ± 0.79	49.89 ± 0.61	2.10 ± 0.04
Random	30%	65.88 ± 0.58	56.80 ± 0.52	1.90 ± 0.03
Random	60%	67.07 ± 1.07	60.78 ± 1.21	1.83 ± 0.55
FAISS	25	67.17 ± 0.65	60.83 ± 0.96	1.84 ± 0.04
All Sib.	100%	68.08 ± 0.76	61.98 ± 0.70	1.78 ± 0.03

Random sampling improves as the negative budget increases: using more negatives leads to higher accuracy and F1, and lower mean geodesic distance. However, even with 60% of the negatives, it remains below the other strategies, suggesting that random sampling needs a large budget to approximate the full negative set. In contrast, FAISS-based selection achieves performance close to the full-memory setting (All Sib.) using only 25 negatives per positive, a small

fraction of the budget required by random sampling, confirming that nearest-neighbor retrieval identifies substantially more informative negatives than uniform sampling.

7.3. JL Dimensionality Ablation

We evaluate the effect of the Johnson–Lindenstrauss target dimension k on classification performance. All experiments are conducted on the COCO validation subset described in Section 6.1 ($\approx 30,000$ samples), with $\lambda = 0$ (always predicting a leaf node) and the complete hierarchy configuration. Results are reported as mean $\pm$ standard deviation over 5 random seeds for the JL projection matrix.

Table 3. Effect of JL target dimension k on classification performance. Higher k reduces distortion at the cost of increased computation. Results are mean $\pm$ std over 5 seeds.

JL dim k	Acc. (%) $\uparrow$	F1 (%) $\uparrow$	MGD $\downarrow$
512	65.37 $\pm$ 0.52	59.62 $\pm$ 0.84	1.95 $\pm$ 0.04
1024	66.51 $\pm$ 1.17	59.83 $\pm$ 0.94	1.88 $\pm$ 0.04
2048	67.30 $\pm$ 0.90	60.12 $\pm$ 0.83	1.84 $\pm$ 0.04
4096	67.99 $\pm$ 0.46	61.75 $\pm$ 0.66	1.79 $\pm$ 0.01
No JL	68.08 $\pm$ 0.76	61.98 $\pm$ 0.70	1.78 $\pm$ 0.03

Performance improves consistently as k increases: accuracy rises from 65.4% at $k = 512$ to 68.0% at $k = 4096$, while MGD decreases from 1.956 to 1.793. The gains diminish as k grows, suggesting a diminishing-returns regime consistent with the JL lemma: larger k reduces the distortion bound ε but the marginal benefit decreases as ε already falls below a threshold where the local neighbourhood structure relevant to EVM fitting is approximately preserved. The variance across seeds also decreases with k , reflecting the improved stability of the projection at higher dimensions.

7.4. Evaluation Results

Figure 1 shows the hierarchical accuracy $\text{HAcc}(\lambda)$ of the proposed method as a function of the open-set threshold λ , compared against the flat baselines on both datasets.

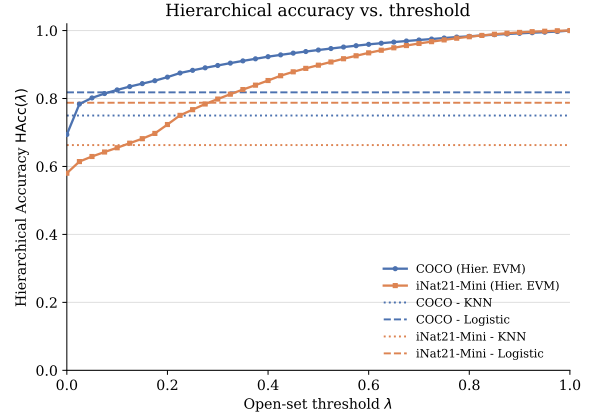


Figure 1. Hierarchical accuracy as a function of the open-set threshold λ on COCO and iNat21-Mini.

COCO. At $\lambda = 0$ (full coverage, always predicting a leaf), the hierarchical EVM achieves $\text{HAcc} = 0.695$, below both the KNN baseline (0.750) and the logistic regression (0.818). However, as λ increases and the model is allowed to abstain at internal nodes, hierarchical accuracy improves rapidly. The model surpasses KNN at $\lambda \approx 0.025$ with a coverage of 0.818, and surpasses the logistic regression at $\lambda \approx 0.100$ with a coverage of 0.707. This indicates that the hierarchical EVM can match or exceed flat classifiers while still providing a specific prediction for approximately 70% of test samples.

iNat21-Mini. The fine-grained nature of iNat21-Mini makes the task substantially harder. At $\lambda = 0$, the method achieves $\text{HAcc} = 0.582$, compared to KNN at 0.663 and logistic regression at 0.787. The model requires a higher threshold to recover: it surpasses KNN at $\lambda \approx 0.125$ (coverage 0.881) and surpasses the logistic regression at $\lambda \approx 0.300$ (coverage 0.652). The gap at full coverage reflects the difficulty of fine-grained discrimination with EVM classifiers on a 10,000-class taxonomy, where class boundaries are inherently less well-separated in the embedding space.

Overall. The results confirm that the hierarchical EVM framework consistently improves in hierarchical accuracy as the threshold increases, at the cost of reduced coverage. On COCO, the model exceeds the logistic regression baseline already at $\lambda \approx 0.10$, while retaining a coverage of $\sim 70.7\%$, and surpasses the KNN baseline at a coverage of $\sim 81.8\%$. On iNat21-Mini, the trade-off follows a similar pattern: the model matches the KNN baseline while retaining a coverage of $\sim 88.1\%$, and the logistic regression baseline at a coverage of $\sim 65.2\%$. The two datasets thus exhibit a qualitatively similar trade-off, with iNat21-Mini requiring a somewhat larger threshold to match its (stronger) logistic

regression baseline, consistent with its substantially larger and deeper label hierarchy (10,000 classes vs. 74).

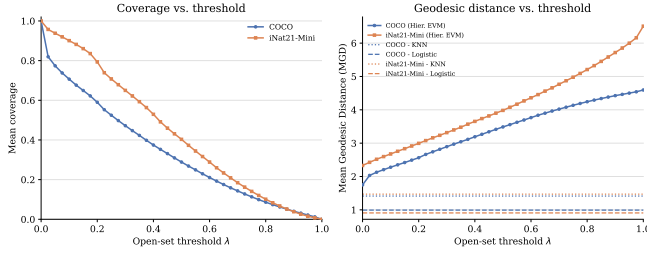


Figure 2. Coverage (left) and Mean Geodesic Distance (right) as a function of the open-set threshold λ on COCO and iNat21-Mini.

Figure 2 shows the evolution of mean coverage and MGD as λ varies. Coverage decreases monotonically on both datasets as the model stops more frequently at internal nodes; COCO drops more steeply due to its deeper and more unbalanced hierarchy. MGD increases with λ on both datasets: internal nodes are farther from the ground-truth leaf in the hierarchy, so stopping earlier raises the average geodesic distance even as hierarchical accuracy improves. Notably, the MGD of the hierarchical EVM remains above that of the flat baselines across all λ values, indicating that the model tends to stop at nodes that are not on the correct ancestral path, rather than abstaining only when uncertain between semantically close siblings. This suggests that the EVM confidence scores are not yet well-calibrated with respect to hierarchical error severity, and points to EVM calibration as a key direction for future work.

8. Limitations

Despite the optimizations introduced, the system still requires all embedding vectors to be stored in memory during training. This limits scalability to larger datasets or higher-dimensional representations.

A current limitation concerns the use of frozen DINOv3 embeddings. Since these representations are primarily organized by visual similarity, they may not align with taxonomic or biological hierarchies whose categories are defined by semantic rather than visual criteria.

A further limitation concerns the calibration of the EVM confidence scores. As shown in the results, MGD increases monotonically with λ even as hierarchical accuracy improves, indicating that the model does not abstain selectively along the correct ancestral path. This suggests that the open-set threshold λ controls prediction specificity but not semantic precision: the model retreats to internal nodes that reduce leaf-level errors without necessarily stopping at the most informative ancestor. Per-node threshold calibration or calibration-aware EVM fitting would be needed to address this.

9. Future work

The current system assumes a *static* hierarchy, fixed before training. A natural extension is an *online* setting where the hierarchy grows over time: starting from a coarse skeleton, the system could insert intermediate nodes, split leaves into finer sub-categories, or add branches as novel concepts are observed, supporting continual adaptation without retraining from scratch. The open-set nature of EVMs already provides a basis for this, since unknown samples can be rejected rather than forced into existing classes. This could also make the hierarchy instance- and context-aware, e.g. refining a generic *cup* node into personalized sub-classes like *white cup* and *blue cup*.

A related direction concerns the mismatch between visual and semantic hierarchies discussed above. Rather than treating all nodes uniformly, future work could label nodes as *visual* or *functional/semantic* and treat them differently, e.g. relying on visual embeddings where category boundaries are visually grounded, and complementary signals (e.g. text-derived embeddings) where they are not.

Overall, this direction points towards hierarchical memory systems that are not only classifiers, but evolving agent-centric models of the visual world.

10. Conclusion

We proposed a hierarchical concept-based classification framework that combines frozen DINOv3 embeddings with a static hierarchy of EVM classifiers. The key property of the framework is its ability to abstain at internal nodes when the visual evidence does not support a fine-grained decision, treating ancestor predictions as meaningful and partially correct rather than simply wrong. This provides a natural mechanism for hierarchical evaluation and a concept-level explanation of each prediction through the traversed path.

Experiments on COCO and iNat21-Mini show that the rejection mechanism is effective: by adjusting the open-set threshold λ , the model consistently improves in hierarchical accuracy at the cost of reduced coverage, surpassing flat baselines at coverage levels that remain practically useful. On COCO, the hierarchical EVM exceeds both KNN and logistic regression while retaining predictions for over 70% of test samples. On the more challenging iNat21-Mini, the gap at full coverage is larger, reflecting the difficulty of fine-grained discrimination across 10,000 species with EVM classifiers; nonetheless, the model still surpasses both flat baselines while retaining coverage of 65% to 88%, showing that the rejection mechanism scales to substantially larger and deeper taxonomies.

The main limitation identified is the calibration of EVM confidence scores with respect to hierarchical error severity: MGD increases with λ even as HAcc improves, indi-

cating that the model does not yet abstain selectively along the correct ancestral path. Addressing this through per-node threshold calibration or calibration-aware EVM fitting represents the most promising direction for future work, alongside extension to an online hierarchy setting where the tree structure can grow and adapt as new concepts are observed.

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DINOv3 embeddings for Hierarchical Concept-Based Classification

Supplementary Material

11. Additional Results

11.1. DinoV3 feasibility

The preliminary comparison shows that replacing the original feature extractor with DINOv3 does not degrade the hierarchical behavior of the pipeline. The hierarchical F-score remains comparable to the original encoder, suggesting that DINOv3 provides a suitable representation space for the instance-based hierarchical classifier.

The geodesic distance results further suggest that DINOv3 produces semantically closer predictions, reducing the severity of errors in the hierarchy. Overall, these results support the use of DINOv3 as an alternative visual backbone.

The following figures report the preliminary results obtained by running the pipeline from *Egocentric Hierarchical Visual Semantics* with DINOv3 as the visual feature extractor.

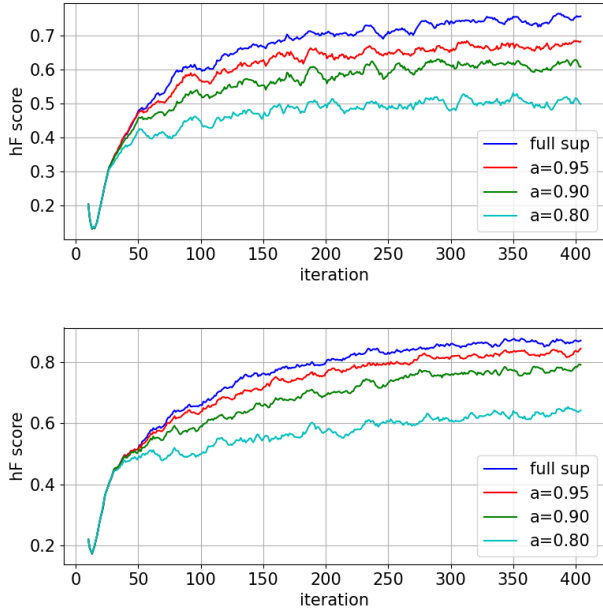


Figure 3. Hierarchical F-score comparison. Top: original feature extractor. Bottom: DINOv3.

The hierarchical F-score obtained with DINOv3 is consistent with the results produced by the original pipeline, suggesting that DINOv3 embeddings can be integrated without degrading the hierarchical classification behavior.

The following figures report the corresponding geodesic distance results.

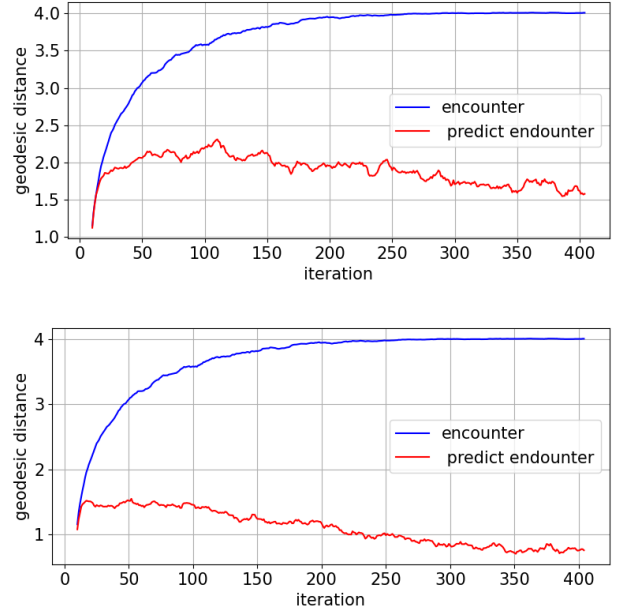


Figure 4. Geodesic distance comparison. Top: original feature extractor. Bottom: DINOv3.

The geodesic distance results show the effect of using DINOv3 embeddings on the semantic distance between predicted and ground-truth nodes.

11.2. Impact of the hierarchical structure

Table 4 reports the results on the 3 hierarchical settings we tested showing that hierarchy depth and complexity does not affect performances significantly.

Table 4. Comparison of hierarchy configurations at $\lambda = 0$, using all siblings. Tested over 5 random seeds.

Hierarchy type	Acc. (%) $\uparrow$	F1 (%) $\uparrow$	MGD $\downarrow$
Complete	68.08 ± 0.76	61.98 ± 0.70	1.78 ± 0.03
Flat	69.17 ± 0.40	63.55 ± 0.98	1.75 ± 0.02
No	69.78 ± 0.66	63.87 ± 0.43	1.79 ± 0.05

12. Dataset Hierarchies

This section reports statistics on the two hierarchies used in our experiments: the COCO hierarchy derived from WordNet, and the biological taxonomy of iNat21-Mini. The *Raw* column refers to the original hierarchy before any pruning, which includes nodes with no associated training examples and nodes with only one child. The *Pruned* column refers

to the hierarchy actually used by the system, obtained by removing empty nodes and collapsing single-child nodes as described in Section 5.

Table 5. Hierarchy statistics before and after pruning.

Metric	COCO		iNat21-Mini	
	Raw	Pruned	Raw	Pruned
Total nodes	9,419	116	16,344	12,445
Leaf nodes	7,573	74	10,000	10,000
Internal nodes	1,846	42	6,344	2,445
Max depth	17	8	7	7
Avg leaf depth	8.86	5.26	7.00	6.52
Avg branching	5.10	2.74	2.58	5.09
Max branching	149	8	216	216

The pruned COCO hierarchy contains 74 leaf nodes, corresponding to the 74 COCO classes present in the dataset after preprocessing, while the remaining 42 internal nodes correspond to intermediate WordNet concepts. The iNat21-Mini hierarchy retains all 10,000 species as leaf nodes; the large reduction in internal nodes (from 6,344 to 2,445) reflects the high proportion of monotypic intermediate ranks in biological nomenclature, which are collapsed during pruning.